

-> 6) Rename operation on Repository

- > If we want to rename remote repository then follow the below steps
Go to your Repository-> settings -> General-> Provide new name-> click on Rename Button.
- > If we want to rename local repository then select repository
and Right click -> Rename

Working on Branches

-> On branch we can perform following operations.

1. We can create new a Branch
2. List out local branch names.
3. We can delete a branch
4. We can clone a branch
5. We can jump from one branch to another branch
6. Push your local branch changes on Central repository branch
7. We can merge one branch into another.

-> Now, we will discuss each operation one by one.

-> Operation: Creating a new Branch

- 1) If we want to create a branch on central Repository then follow the below steps
Select your repository-> Click on "Branches" Tab -> New Branch
- 2) If want to create a branch on local repository then you can use
either "`git branch`" or "`git checkout`" command
 - 1) Creating a branch using "`git branch`" command
`git branch` command is used for different purpose but If you want to create a branch
then follow the below steps
 - i) First switch to parent branch by using "`git switch parent-branch-name`" command
or "`git checkout parent-branch-name`" command
 - ii) Create a branch using "`git branch new-branch-name`" command.

Q. How to use "git checkout" command to create a new branch?

-> If we want to create a branch by using "git checkout" command then follow the below steps

Step 1) Switch to parent branch by using "git switch" or "git checkout" command

"git checkout parent-branch-name"

or "git switch parent-branch-name"

Step 2: Create new branch by using "git checkout -b new-branch-name"

Here this command is performing two activity at a time.

1) Create a new branch with given name

2) Switch to newly created branch.

Q. What is difference between below commands syntax?

1) git branch new-branch-name

2) git checkout -b new-branch-name

-> First syntax just create new branch from current branch but second syntax creating new branch from current branch and switching to newly creating branch

Conclusion about creating a branch

1) First switch to parent branch

2) Create your new branch

Q. How to cross check parent branch name?

-> If you want to cross check from which parent branch you have created your branch then use "git reflog"

Syntax: git reflog show your_branch_name

In the output look for the checkout entries that show moving from one branch to another.

Q. We have three command to create a new branch, these three command syntax is as below

1) git branch new-branch-name

2) git checkout -b new-branch-name (here -b is stands for branch)

3) git switch -c new-branch-name (here -c stands for create)

among above syntax which syntax is recommended?

-> From the above syntax below two syntaxes are recommended

1) git branch new-branch-name

2) git checkout -b new-branch-name (here -b is stands for branch)

Q. While creating a branch on local repository which precaution you should take?

-> by default new branch get created on current branch so whenever we want to create a branch then first switch to parent branch and then create a new branch

-> Operation: Listing out branches.

- 1) If we want to display local branch names then just type "git branch" and press enter. You will get list of all local branches
- 2) If you want to check central repository branch names then follow below steps.
 - 1) Select your repository -> Click Branches, We will get overview of all branches.
 - 2) If you want to check yours branches then click on "yours" tab
 - 3) If you want to check active branches then click on "Active" tab
 - 4) If you want to check stale branches then click on "Stale" tab
 - 5) If you want to check all branches then click on "All" tab

-> Operation: Deleting a branch

- 1) If you want to delete a branch from local repository then use "git branch" command with -d option.

Syntax: git branch -d branch-name

* If we want to delete multiple branches at a time then use below syntax

Syntax: git branch -d first_branch_name second_branch_name

Note: Branch names should be seperated by space

- 2) If you want to delete remote repository branch then follow the steps

Select Repository -> Click on "Branches" Tab -> Search Branch-> select Branch-> right side we have delete icon,click click on that.

-> Operation: Cloneing a branch

-> When you are going to clone a branch there may be two possiblities

- 1) Repository is not available on local System.
- 2) Repository is already available on local system

Possibility 1: Remote Repository is not available on local system.

If repository is not available on local system and you want to clone a branch then use below command.

Syntax: git clone -b "branch-name" "repository_url"

-> Possibility 2: Repository is already available on local system

-> Remote Repository is available means either main branch is available or any other branch is available on local system.

-> If Git repository already available on local system and you want to clone a specific branch of that repository then you don't need to clone the entire repository again. Instead, just clone your required branch.

-> Steps to clone required branch

Step 1) First cross check all branches(local and remote) by using below command.

`git branch -a` or `git branch --all`

Note 1: Local branches: Those already checked out on your system.

Note 2: Remote branches: Branches available on the remote repository but not on local(remote branch name start with remote)

Step 2) clone your newly create remote branch on local system by using "git checkout" command

Syntax: `git checkout -b branch-name origin/branch-name` (here use /)

Above command "git checkout" does two things.

1. -b branch-name: Creates a new local branch with given name.

2. origin/branch-name: Tracks specified the remote branch from the origin remote.

Step 3) Execute the "git pull" to get Latest Changes but it is optional

Operation: Jumping from one branch to another.

-> If we want to jump from one branch to another branch then use

"git checkout" or "git switch" command in below format

Syntax of git checkout: `git checkout branch-name`

Syntax of git switch : `git switch branch-name`

Note: When are switching from one branch to another branch then comit or stash your current branch changes otherwise we will get an error.

Operation: push local branch changes on remote branch

-> open git bash in your local repository-> switch your desired branch -> simply type "git push" and press enter.